

ORIGINAL ARTICLE

WILEY

The curse of knowledge when teaching statistics

Itamar Shatz 

Department of Theoretical and Applied Linguistics, University of Cambridge, Cambridge, UK

Correspondence

Itamar Shatz, Department of Theoretical and Applied Linguistics, Raised Faculty Building, University of Cambridge, Sidgwick Avenue, Cambridge, CB3 9DA, UK.

Email: is442@cam.ac.uk

Abstract

When teaching statistics, educators sometimes overestimate their students' knowledge and abilities. This is due to the *curse of knowledge*, a cognitive bias that causes people—especially experts—to overestimate how likely others are to know and understand the same things as them. This can lead to various issues, including struggling to communicate with students, and making students feel less comfortable in the classroom. To address this, educators should first identify situations where this bias can affect their teaching. In doing so, they should consider relevant risk factors, and potentially also solicit feedback from relevant individuals. Then, educators can reduce this bias and its impact on their teaching by using techniques such as keeping the curse of knowledge and their audience in mind, assessing students' knowledge, assuming lack of knowledge unless there is strong evidence to the contrary, and avoiding saying that things are obvious.

KEYWORDS

cognitive bias, communication, curse of knowledge, debiasing, education, egocentric bias, teaching statistics

1 | INTRODUCTION

People—and especially experts—often struggle to properly understand the perspective of those who do not know as much as them. This is due to a cognitive bias known as the *curse of knowledge* (or sometimes as the *curse of expertise*) [4,5].

An example of the curse of knowledge appears in Figure 1, where two experts are having a discussion about how much the average person knows regarding their field. They wildly overestimate this, despite understanding that people outside their field likely know less about it than they—the experts—do.

If you the reader—like most people—are not an expert in geochemistry, then you likely have no idea what these experts are talking about, even though it feels

obvious to them that the average person is familiar with the concepts that they are discussing. An equivalent comic for statistics could involve experts saying something like “The average person probably knows only a few statistical tests, like the t-test and ANOVA, and maybe also the chi-squared test.” Similarly to the geochemistry example, even though knowledge of these tests may seem obvious to those who teach and use statistics, the average person is likely unaware of these tests, or even of what a statistical test is.

Essentially, the curse of knowledge causes those with certain knowledge—and especially expertise in a field—to overestimate how likely others are to know and understand the same things as them. This is a problem that educators frequently encounter, since they are generally much more knowledgeable about the subjects that they

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2022 The Author. *Teaching Statistics* published by John Wiley & Sons Ltd on behalf of Teaching Statistics Trust.

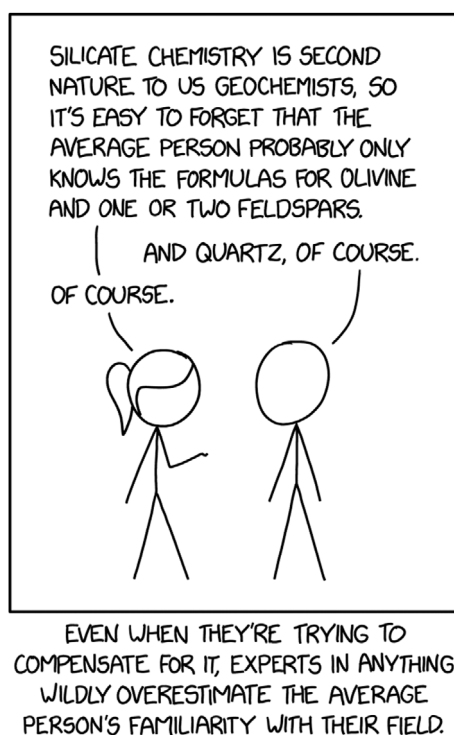


FIGURE 1 A comic illustrating a typical example of experts displaying the curse of knowledge. This comic, titled “Average Familiarity”, comes with the following added caption: “How could anyone consider themselves a well-rounded adult without a basic understanding of silicate geochemistry? Silicates are everywhere! It’s hard to throw a rock without throwing one!”. It is from <https://xkcd.com/2501/>, and is available under a CC BY-NC 2.5 License (<https://xkcd.com/license.html>)

teach than their students are. For example, a statistics educator who is well-familiar with the concept of p -value may assume that various aspects of it—such as that it is not an effect size—are obvious to their students, even when that is not the case.

This paper presents an expository overview of the curse of knowledge, positioned in the context of statistics education. Its goal is to help statistics educators understand this bias and avoid it in their teaching. In addition, the material presented here may also help educators in other fields, and may also help certain practitioners who are not educators (eg, science communicators).

2 | THE CURSE OF KNOWLEDGE IN STATISTICS EDUCATION

The curse of knowledge can cause various issues in statistics education. For example, it may cause a teacher to skip explaining something (eg, that the word “average” may have more than one meaning) if they take it for granted that their students already know this. Similarly, the curse

of knowledge can cause a teacher to overestimate how fast and how well students will understand the relationship between standard errors and confidence intervals, and consequently to explain it too quickly for them to follow.

Such issues, in turn, can reduce the effectiveness of teaching. For example, this can happen if students are taught advanced methods without sufficient checking of their understanding of underlying statistical fundamentals, and consequently misuse those methods (eg, if they are taught how to use ANOVA, without sufficient understanding of the p -value that it outputs).

Furthermore, such issues can also interfere with the learning environment in other ways. For example, they can make the classroom feel less comfortable and inclusive, if they make students feel that the teacher does not understand them and does not show empathy, or if they make students feel stupid since they are unfamiliar with concepts the teacher thinks are obvious.

In addition, the curse of knowledge can cause similar issues in teaching-related contexts outside the classroom, and especially those involving interpersonal communication. For example, the curse of knowledge can make it hard for a statistics instructor to figure out how to explain the central limit theorem to a struggling student under time pressure that hinders the instructor's ability to understand the student's perspective. Similarly, the curse of knowledge can make it hard for a statistics instructor to give an effective presentation about statistical methods to non-statistics peers (eg, teachers in other subjects) or to a lay audience, again due to a difficulty in understanding their perspectives.

However, there are two important caveats about these issues.

First, although the curse of knowledge *may* influence the way educators teach and communicate statistics, this does not mean that it always does so. For example, there are cases where educators use techniques such as reflection—which we will see later—to debias their thinking, and therefore eliminate the influence of the curse of knowledge on their teaching.

Second, the curse of knowledge can play a role not only in statistics education, but also in the teaching of other subjects—such as biology, psychology, and computer science—given that there is a gap in knowledge between teachers and their students in those fields too. Nevertheless, it seems reasonable to suggest that the curse of knowledge likely plays a bigger role in statistics education than in the education of many other fields, given the high interdisciplinary applicability of statistics.

Specifically, a key reason for this is that statistics experts often teach statistics to students whose background is very different from their own (eg, biology students), which can increase the gap in knowledge and

perspective between them and their students (eg, as opposed to a biology professor teaching biology students). Furthermore, students' exposure to statistics is often unsystematic and non-standardized, as it is driven primarily by what they have been exposed to or taught in their educational background, and in their current discipline or study area. In contrast, statistics educators, who should have received a thorough and extensive education in statistics, need to do the work to discover their students' backgrounds and understand their students' perspectives.

Future research could determine whether there is indeed such a difference in the curse of knowledge across the teaching of different fields, by investigating how different teachers' perception of students' knowledge is from students' actual knowledge across fields. Such research could also investigate other aspects of the curse of knowledge in statistics education, such as whether certain concepts in statistics (eg, p -value and estimation) tend to involve a greater curse of knowledge than others, and if so, whether this is moderated by factors such as students' educational background and level (eg, high school vs college). Determining this could help statistics educators identify cases where they are more likely to display the curse of knowledge, which could help them avoid it more effectively.

3 | PSYCHOLOGY AND CAUSES OF THE CURSE OF KNOWLEDGE

The curse of knowledge is generally attributed to two main cognitive mechanisms, both of which can influence people simultaneously [4].

The first mechanism is *inhibitory control*; this represents people's difficulty to ignore information that they possess when trying to understand other people's perspectives. For example, this means that if a statistics teacher knows a certain concept, such as linear regression, then it can be hard for them to ignore their own knowledge of this concept when trying to figure out the thought process of students who have not yet learned about it.

The second mechanism is *fluency misattribution*; this represents people's tendency to overestimate the likelihood that others have the same information as them, as well as the ease with which others can process this information. For example, this means that if a teacher knows how to use a certain formula, such as the one for calculating standard deviation (SD), then they might overestimate the likelihood that their students know how to use this formula too, and underestimate the time it will take them to apply, or become familiar with, it.

In addition, other cognitive mechanisms can also play a role in causing the curse of knowledge. This includes, for example, the *anchoring-and-adjustment process*, which occurs when people remain anchored to their own perspective when trying to reason about someone else's perspective [4]. This also includes having issues with one's *theory of mind*, which is the ability to understand that others can be different from oneself when it comes to things such as perceptions, thoughts, and emotions [2,7].

Finally, from a psychological perspective, the curse of knowledge is similar to the concept of *epistemic egocentrism* [4,10], and is considered to be a type of *egocentric bias*, since it causes people to rely too heavily on their own perspective when trying to reason about other people's perspectives [6,11]. However, a unique feature of the curse of knowledge is its asymmetry, since it influences only those who attempt to reason about the perspective of those who have less knowledge, but not the other way around [1,3].

4 | DEALING WITH THE CURSE OF KNOWLEDGE

Now that we have seen what the curse of knowledge is, why it is a problem, and what causes it, a key remaining question is what we can do to deal with it effectively.

Below, we will see techniques that can help with this. Many statistics educators are likely already familiar with at least some of these techniques, and use them to improve their teaching, including to mitigate the curse of knowledge and other biases. Accordingly, the goals of this section are to highlight at least some of these techniques, to potentially help all of us, no matter how familiar we are with these techniques, to use them more effectively, particularly in the context of the curse of knowledge.

As statistics educators looking to reduce the curse of knowledge, the first thing we can do after learning about this bias is reflect on our teaching, in order to identify situations where we may have displayed this bias in the past or are likely to display it in the future. When doing this, we should focus on situations where we are particularly likely to display this bias, due to factors such as:

1. Having an especially large gap in knowledge between us and our students, for example, if we are a statistics professor who is giving a talk to middle-school students.
2. Engaging with diverse audiences—that have very different statistics knowledge—over short periods of

time, for example, if we immediately go from teaching statistics students to teaching humanities students.

3. Having recently gone over the learning material extensively—for example, as part of lesson preparation—which makes it much fresher in our mind.

When reflecting on teaching, it is important to also keep in mind the *bias blind spot*, which is the tendency to underestimate and overlook our own biases [8,9,12]. This blind spot means that we are prone to underestimating and ignoring our curse of knowledge, so even if we intuitively feel that we do not display this bias, it will likely be worthwhile to engage in reflection. In addition, it may be beneficial to solicit relevant feedback from others, for example, by asking students if there are concepts they would have liked to have seen or heard explanations in more detail.

Once someone better understands when and how they are likely to display the curse of knowledge, they can move on to debiasing their teaching accordingly. The following are several key techniques that can be used to achieve this:

1. **Keep the curse of knowledge in mind while teaching and preparing.** For example, if I catch myself thinking “it’s obvious that the p -value isn’t an effect size, so I don’t need to mention it”, I can stop and ask myself “could I be overestimating how obvious this is to my students, just because it’s so obvious to me?”.
2. **Always have the audience in mind.** For example, in preparing a lesson plan and wondering whether there is a need to explain a certain concept (eg, SD), one can ask oneself what level the students are at, how likely they are to know this concept, and how likely they are to be able to understand this concept without an explanation. When doing this, it must be remembered that even within the same classroom, the students are likely to have very different knowledge and abilities.
3. **Assess students’ knowledge.** For example, one can look at recent student work to identify concepts that students are struggling with despite what a teacher might intuitively expect. In addition, while teaching, one can actively ask students whether they understand certain concepts, or whether they would like more or repeated explanation of those concepts.
4. **When in doubt, assume lack of knowledge.** For example, if one is unsure whether students are familiar with a certain concept, it is often better to explain it than not. This is because failing to explain necessary

concepts can lead to substantial issues—such as students not understanding what is going on and consequently feeling uncomfortable—which generally outweigh any potential downsides associated with explaining concepts that students already know. However, it can be a significant turnoff, and lead to losing students’ cooperation, if time is wasted going over familiar ground. Being open and honest with students, as well as learning their backgrounds, can achieve the desirable balance.

5. **Avoid saying that things are obvious.** For example, one should generally avoid saying things such as “*obviously* this is wrong”, “*of course* that’s not the case”, or “as this equation *clearly* shows”. This is because what is being discussed may not be as obvious to students as may be thought, and by implying that it should be obvious one can make students feel that there is something wrong with them.

5 | CONCLUSIONS

A common mistake that educators make while teaching statistics is assuming that their students know more than they really do. This is often due to a cognitive bias called the *curse of knowledge*, which causes people—and especially experts—to overestimate how likely others are to know and understand the same things as them. This can lead to various issues, including struggling to communicate with students, and making students feel less comfortable in the classroom.

To deal with the curse of knowledge, one should first identify situations where it is likely to affect one’s teaching. When doing this, one should consider factors that increase the risk of this bias, such as having an especially large gap in knowledge from one’s students. Feedback can also be solicited from relevant individuals, such as peers and students.

Then, this issue and its influence on teaching can be mitigated by using various techniques. These include keeping the curse of knowledge and the audience in mind, assessing students’ knowledge, assuming lack of knowledge unless there is strong evidence to the contrary, being open with students, and avoiding saying that things are obvious.

Doing this can help understand the range of students, communicate with them better, teach them more effectively, and make them feel more comfortable in the learning environment.

ORCID

Itamar Shatz  <https://orcid.org/0000-0001-8916-9010>

REFERENCES

1. K. Bhandari and H. Barth, *Show or tell: Testimony is sufficient to induce the curse of knowledge in three- and four-year-olds*, *Quart. J. Exp. Psychol.* **63** (2010), no. 2, 209–215. <https://doi.org/10.1080/17470210903168250>.
2. S. A. J. Birch, *When knowledge is a curse: Children's and adults' reasoning about mental states*, *Curr. Direct. Psychol. Sci.* **14** (2005), no. 1, 25–29. <https://doi.org/10.1111/j.0963-7214.2005.00328.x>.
3. S. A. J. Birch and P. Bloom, *Children are cursed: An asymmetric bias in mental-state attribution*, *Psychol. Sci.* **14** (2003), no. 3, 283–286. <https://doi.org/10.1111/1467-9280.03436>.
4. S. A. J. Birch, P. E. Brosseau-Liard, T. Haddock, and S. E. Ghrear, *A 'curse of knowledge' in the absence of knowledge? People misattribute fluency when judging how common knowledge is among their peers*, *Cognition* **166** (2017), 447–458. <https://doi.org/10.1016/j.cognition.2017.04.015>.
5. C. Camerer, G. Loewenstein, and M. Weber, *The curse of knowledge in economic settings: An experimental analysis*, *J. Politic. Econ.* **97** (1989), no. 5, 1232–1254. <https://doi.org/10.1086/261651>.
6. B. G. Farrar and L. Ostojić, *Does social distance modulate adults' egocentric biases when reasoning about false beliefs?* *PLoS One* **13** (2018), no. 6, 1–17. <https://doi.org/10.1371/journal.pone.0198616>.
7. D. Massaro, I. Castelli, L. Sanvito, and A. Marchetti, *The 'I knew it all along' phenomenon: Second-order false belief understanding and the curse of knowledge in primary school children*, *Eur. J. Psychol. Educ.* **29** (2014), no. 3, 311–326. <https://doi.org/10.1007/s10212-013-0200-7>.
8. E. Pronin and M. B. Kugler, *Valuing thoughts, ignoring behavior: The introspection illusion as a source of the bias blind spot*, *J. Exp. Soc. Psychol.* **43** (2007), no. 4, 565–578. <https://doi.org/10.1016/j.jesp.2006.05.011>.
9. E. Pronin, D. Y. Lin, and L. Ross, *The bias blind spot: Perceptions of bias in self versus others*, *Personal. Soc. Psychol. Bull.* **28** (2002), no. 3, 369–381. <https://doi.org/10.1177/0146167202286008>.
10. E. B. Royzman, K. W. Cassidy, and J. Baron, *"I know, you know": Epistemic egocentrism in children and adults*, *Rev. Gen. Psychol.* **7** (2003), no. 1, 38–65. <https://doi.org/10.1037/1089-2680.7.1.38>.
11. R. A. Ryskin and S. Brown-Schmidt, *Do adults show a curse of knowledge in false-belief reasoning? A robust estimate of the true effect size*, *PLoS One* **9** (2014), no. 3, 1–8. <https://doi.org/10.1371/journal.pone.0092406>.
12. I. Scopelliti, C. K. Morewedge, E. McCormick, H. L. Min, S. Lebrecht, and K. S. Kassam, *Bias blind spot: Structure, measurement, and consequences*, *Manag. Sci.* **61** (2015), no. 10, 2468–2486. <https://doi.org/10.1287/mnsc.2014.2096>.

How to cite this article: I. Shatz, *The curse of knowledge when teaching statistics*, *Teach. Stat.* **45** (2023), 22–26. <https://doi.org/10.1111/test.12320>